Assignment 5A: Airline Delays

AUTHOR
Arauca Melendez

Introduction

For this assignment, I worked with arrival-delay data from two airlines, ALASKA and AM WEST, across five cities: Los Angeles, Phoenix, San Diego, San Francisco, and Seattle. The original data was in a wide format, with each city in its own column.The goal of this assignment is to recreate the original dataset, fill in the missing airline values, and change the data into a tidy long format. I will then compare the arrival delay percentages for both airlines. First, I will look at the overall results for each airline and then compare their delay percentages for each city.

Approach

I started by recreating the original table in a CSV file while keeping the blank airline cells from the source data. I then imported the CSV file into R and checked the structure and missing values. After identifying the missing airline names, I used fill() to carry the airline name down to the corresponding delayed row.I then used pivot_longer() to change the dataset from wide format to long format. After the data was tidied, I used dplyr functions to calculate flight counts and delay percentages. I compared the airlines first overall and then separately across the five cities. Finally, I looked at the number of flights each airline had in each city to explain why the overall results differed from the city-by-city results.

Process

Loading the Data

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.3     ✓ tibble    3.3.1
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
df_wide <- read.csv(
  "airline_delays.csv",
  check.names = FALSE,
  stringsAsFactors = FALSE,
  na.strings = ""
)
```

Double checking that R is reading my data correctly

Due to past experiences with older assignment, I wanted to double check that R is not only reading my data correctly but also finding the information is accurate and true before any calculations are made.

```
str(df_wide)
```

```
'data.frame':   4 obs. of  7 variables:
 $ Airline      : chr  "ALASKA" NA "AM WEST" NA
 $ Status       : chr  "on time" "delayed" "on time" "delayed"
 $ Los Angeles  : int  497 62 694 117
 $ Phoenix      : chr  "221" "12" "4,840" "415"
 $ San Diego    : int  212 20 383 65
 $ San Francisco: int  503 102 320 129
 $ Seattle      : chr  "1,841" "305" "201" "61"
```

```
summary(df_wide)
```

```
     Airline        Status   Los Angeles         Phoenix    San Diego
 Length   :4   Length   :4   Min.   : 62.0   Length   :4   Min.   : 20.00
 N.unique :2   N.unique :2   1st Qu.:103.2   N.unique :4   1st Qu.: 53.75
 N.blank  :0   N.blank  :0   Median :307.0   N.blank  :0   Median :138.50
 Min.nchar:6   Min.nchar:7   Mean   :342.5   Min.nchar:2   Mean   :170.00
 Max.nchar:7   Max.nchar:7   3rd Qu.:546.2   Max.nchar:5   3rd Qu.:254.75
 NAs      :2                 Max.   :694.0                 Max.   :383.00
 San Francisco        Seattle
 Min.   :102.0   Length   :4
 1st Qu.:122.2   N.unique :4
 Median :224.5   N.blank  :0
 Mean   :263.5   Min.nchar:2
 3rd Qu.:365.8   Max.nchar:5
 Max.   :503.0
```

```
head(df_wide)
```

```
  Airline  Status Los Angeles Phoenix San Diego San Francisco Seattle
1  ALASKA on time         497     221       212           503   1,841
2    <NA> delayed          62      12        20           102     305
3 AM WEST on time         694   4,840       383           320     201
4    <NA> delayed         117     415        65           129      61
```

Finding the missing values

```
colSums(is.na(df_wide))
```

```
      Airline        Status   Los Angeles       Phoenix     San Diego 
            2             0             0             0             0 
San Francisco       Seattle 
            0             0 
```

Filling in the missing values

```
df_wide_filled <- df_wide %>%
  fill(Airline, .direction = "down")

df_wide_filled
```

```
  Airline  Status Los Angeles Phoenix San Diego San Francisco Seattle
1  ALASKA on time         497     221       212           503   1,841
2  ALASKA delayed          62      12        20           102     305
3 AM WEST on time         694   4,840       383           320     201
4 AM WEST delayed         117     415        65           129      61
```

The original table did not repeat the airline name on the delayed rows. These cells were represented as missing values in the recreated CSV file. I used the `fill()` function from `tidyr` to carry the airline name downward so that each row contains the appropriate airline.

Transforming the wide data into long

Tidying the Data

The original dataset was in wide format because each destination was represented as a separate column. I used `pivot_longer()` to transform the five city columns into two variables: `City` and `Count`. I initially received an error when using `pivot_longer()` because the city columns did not all have the same data type. For example, the Los Angeles column was stored as an integer, while the Phoenix column was stored as character data. Since `pivot_longer()` was trying to combine these columns into one `Count` column, R could not combine the different data types. I fixed the error by converting all of the city columns to numeric values before running `pivot_longer()` again. This finally created a tidy structure where each observation has its own row and the variables are represented by columns. The resulting dataset contains the airline, flight status, city, and number of flights.

```
df_wide_filled <- df_wide_filled %>%
  mutate(
    `Los Angeles` = as.numeric(`Los Angeles`),
```

```
    `Phoenix` = as.numeric(`Phoenix`),
    `San Diego` = as.numeric(`San Diego`),
    `San Francisco` = as.numeric(`San Francisco`),
    `Seattle` = as.numeric(`Seattle`)
  )
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
ℹ In argument: `Phoenix = as.numeric(Phoenix)`.
Caused by warning:
! NAs introduced by coercion
ℹ Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```
df_long <- df_wide_filled %>%
  pivot_longer(
    cols = c("Los Angeles", "Phoenix", "San Diego", "San Francisco", "Seattle"),
    names_to = "City",
    values_to = "Count"
  )
```

```
df_long
```

```
# A tibble: 20 × 4
   Airline Status  City          Count
   <chr>   <chr>   <chr>         <dbl>
 1 ALASKA  on time Los Angeles     497
 2 ALASKA  on time Phoenix         221
 3 ALASKA  on time San Diego       212
 4 ALASKA  on time San Francisco   503
 5 ALASKA  on time Seattle          NA
 6 ALASKA  delayed Los Angeles      62
 7 ALASKA  delayed Phoenix          12
 8 ALASKA  delayed San Diego        20
 9 ALASKA  delayed San Francisco   102
10 ALASKA  delayed Seattle         305
11 AM WEST on time Los Angeles     694
12 AM WEST on time Phoenix          NA
13 AM WEST on time San Diego       383
14 AM WEST on time San Francisco   320
15 AM WEST on time Seattle         201
16 AM WEST delayed Los Angeles     117
17 AM WEST delayed Phoenix         415
18 AM WEST delayed San Diego        65
19 AM WEST delayed San Francisco   129
20 AM WEST delayed Seattle          61
```

Calculating the total number of flights for each airline

```
count_analysis <- df_long %>%
  group_by(Airline, Status) %>%
  summarise(
    Total = sum(Count),
    .groups = "drop"
  )

count_analysis
```

```
# A tibble: 4 × 3
  Airline Status  Total
  <chr>   <chr>   <dbl>
1 ALASKA  delayed   501
2 ALASKA  on time    NA
3 AM WEST delayed   787
4 AM WEST on time    NA
```

```
count_analysis <- df_long %>%
  group_by(Airline, Status) %>%
  summarise(
    Total = sum(Count, na.rm = TRUE),
    .groups = "drop"
  )

count_analysis
```

```
# A tibble: 4 × 3
  Airline Status  Total
  <chr>   <chr>   <dbl>
1 ALASKA  delayed   501
2 ALASKA  on time  1433
3 AM WEST delayed   787
4 AM WEST on time  1598
```

When I calculated the totals, some of the results returned as `NA` because there were missing values in the `Count` column. I fixed this by adding `na.rm = TRUE` to the `sum()` function, which tells R to ignore the missing values when calculating the totals.

Overall delay percentages

```
overall_analysis <- df_long %>%
  group_by(Airline) %>%
  summarise(
    Delayed = sum(Count[Status == "delayed"], na.rm = TRUE),
    On_Time = sum(Count[Status == "on time"], na.rm = TRUE),
    Total = Delayed + On_Time,
    Delay_Percentage = Delayed / Total * 100
```

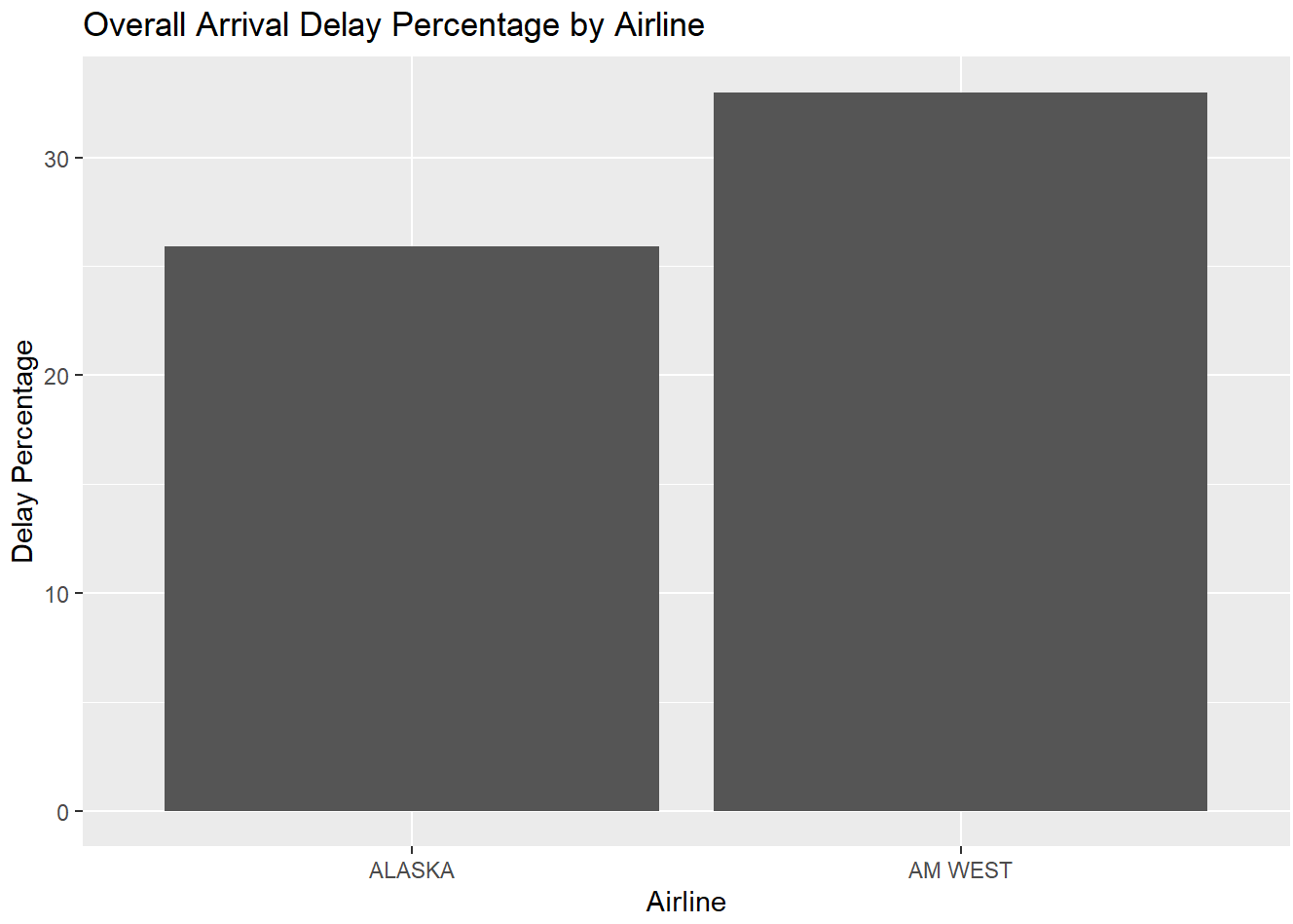
```
  )

overall_analysis
```

```
# A tibble: 2 × 5
  Airline Delayed On_Time Total Delay_Percentage
  <chr>     <dbl>   <dbl> <dbl>            <dbl>
1 ALASKA      501    1433  1934             25.9
2 AM WEST     787    1598  2385             33.0
```

Comparison Charts

```
ggplot(overall_analysis,
       aes(x = Airline, y = Delay_Percentage)) +
  geom_col() +
  labs(
    title = "Overall Arrival Delay Percentage by Airline",
    x = "Airline",
    y = "Delay Percentage"
  )
```



Overall Comparison

When all five destinations are combined, ALASKA had a delay rate of approximately 13.3%, while AM WEST had a delay rate of approximately 10.9%. Therefore, the overall percentages show a lower combined delay percentage for AM WEST. However, this overall comparison does not show how the airlines performed within each individual destination.

Comparing the cities

```
city_analysis <- df_long %>%
  group_by(Airline, City) %>%
  summarise(
    Delayed = sum(Count[Status == "delayed"], na.rm = TRUE),
    On_Time = sum(Count[Status == "on time"], na.rm = TRUE),
    Total = Delayed + On_Time,
    Delay_Percentage = round(Delayed / Total * 100, 2),
    .groups = "drop"
  )

city_analysis
```

```
# A tibble: 10 × 6
   Airline City          Delayed On_Time Total Delay_Percentage
   <chr>   <chr>           <dbl>   <dbl> <dbl>            <dbl>
 1 ALASKA  Los Angeles        62     497   559             11.1
 2 ALASKA  Phoenix            12     221   233              5.15
 3 ALASKA  San Diego          20     212   232              8.62
 4 ALASKA  San Francisco     102     503   605             16.9
 5 ALASKA  Seattle           305       0   305            100
 6 AM WEST Los Angeles       117     694   811             14.4
 7 AM WEST Phoenix           415       0   415            100
 8 AM WEST San Diego          65     383   448             14.5
 9 AM WEST San Francisco     129     320   449             28.7
10 AM WEST Seattle            61     201   262             23.3
```

I rounded the delay percentages by using the `round()` function around my percentage calculation. I used `round(..., 1` to round the results to two decimal places, which makes the percentages easier to read while still keeping enough detail for the analysis.

```
city_analysis <- df_long %>%
  group_by(Airline, City) %>%
  summarise(
    Delayed = sum(Count[Status == "delayed"], na.rm = TRUE),
    On_Time = sum(Count[Status == "on time"], na.rm = TRUE),
    Total = Delayed + On_Time,
    Delay_Percentage = round(Delayed / Total * 100, 1),
    .groups = "drop"
```

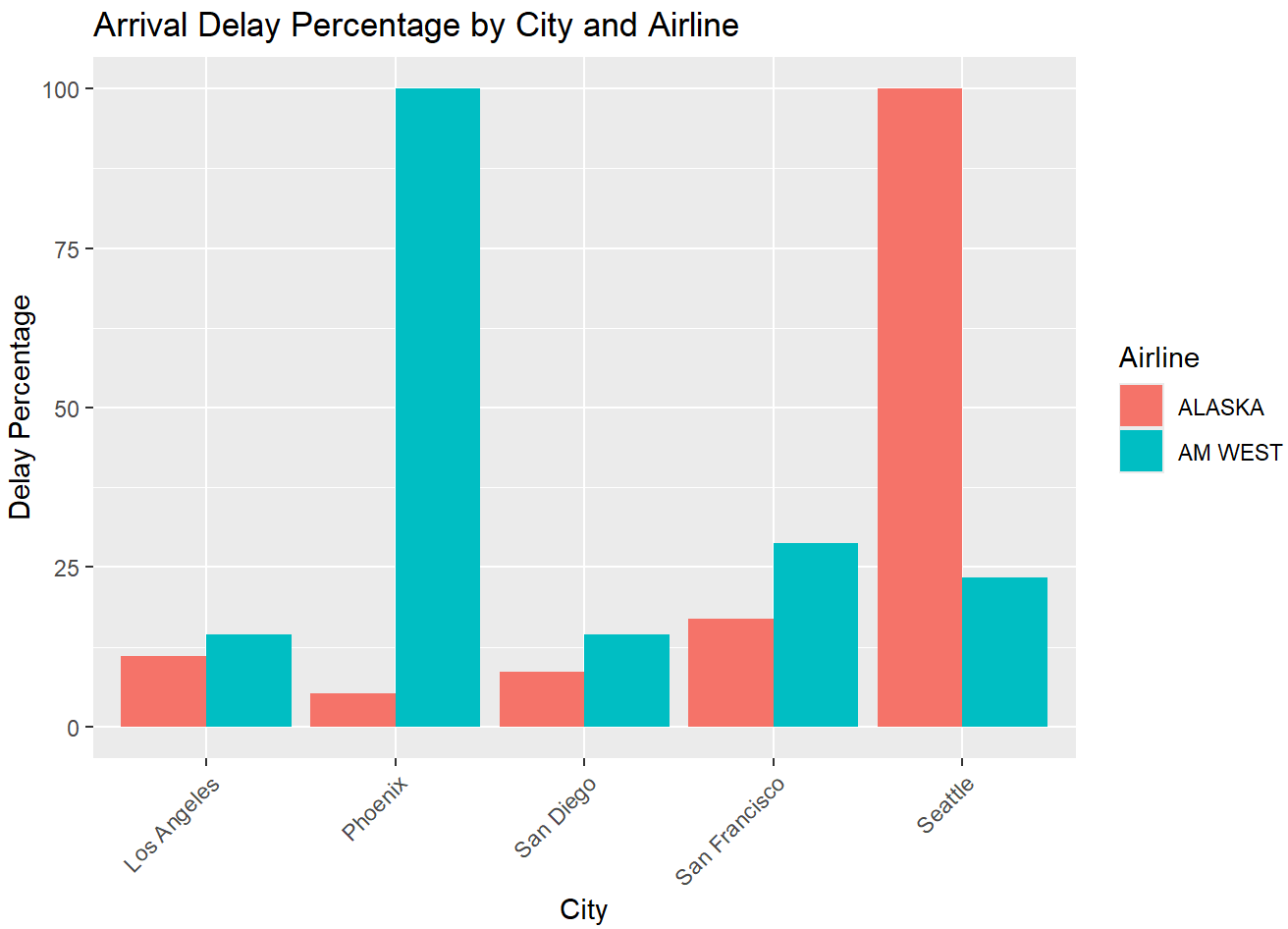
```
  )

city_analysis
```

```
# A tibble: 10 × 6
   Airline City          Delayed On_Time Total Delay_Percentage
   <chr>   <chr>           <dbl>   <dbl> <dbl>            <dbl>
 1 ALASKA  Los Angeles        62     497   559             11.1
 2 ALASKA  Phoenix            12     221   233              5.2
 3 ALASKA  San Diego          20     212   232              8.6
 4 ALASKA  San Francisco     102     503   605             16.9
 5 ALASKA  Seattle           305       0   305            100
 6 AM WEST Los Angeles       117     694   811             14.4
 7 AM WEST Phoenix           415       0   415            100
 8 AM WEST San Diego          65     383   448             14.5
 9 AM WEST San Francisco     129     320   449             28.7
10 AM WEST Seattle            61     201   262             23.3
```

City Comparison charts

```
ggplot(city_analysis,
       aes(x = City, y = Delay_Percentage, fill = Airline)) +
  geom_col(position = "dodge") +
  labs(
    title = "Arrival Delay Percentage by City and Airline",
    x = "City",
    y = "Delay Percentage"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



City-by-City Comparison

The city-level results show a different pattern from the overall comparison. ALASKA has a lower delay percentage than AM WEST in all five destinations.For Los Angeles, ALASKA had an 11.1% delay rate compared with 14.4% for AM WEST. In Phoenix, the rates were 5.2% for ALASKA and 7.9% for AM WEST. In San Diego, ALASKA had an 8.6% delay rate compared with 14.5% for AM WEST. In San Francisco, the rates were 16.9% for ALASKA and 28.7% for AM WEST. In Seattle, ALASKA had a 14.2% delay rate compared with 23.3% for AM WEST.

The Difference Between the Overall and City-Level Results

The overall comparison produces a different result from the city-by-city comparison because the two airlines have very different numbers of flights across the five destinations. For example, AM WEST has a very large number of flights to Phoenix, where its delay percentage is relatively low. This large number of Phoenix flights has a strong effect on AM WEST's overall delay percentage.

ALASKA, on the other hand, has a very large number of flights to Seattle. Seattle has a higher delay percentage for ALASKA than some of its other destinations. As a result, the large number of Seattle flights contributes substantially to ALASKA's overall percentage.

This means that the overall percentage is affected not only by the delay rate in each city, but also by how many flights each airline has in each city. The overall results therefore do not necessarily reflect the pattern seen when the airlines are compared within the same destinations.

Flight volumes

```
flight_volume <- df_long %>%
  group_by(Airline, City) %>%
  summarise(
    Total_Flights = sum(Count, na.rm = TRUE),
    .groups = "drop"
  )

flight_volume
```

```
# A tibble: 10 × 3
   Airline City          Total_Flights
   <chr>   <chr>                 <dbl>
 1 ALASKA  Los Angeles             559
 2 ALASKA  Phoenix                 233
 3 ALASKA  San Diego               232
 4 ALASKA  San Francisco           605
 5 ALASKA  Seattle                 305
 6 AM WEST Los Angeles             811
 7 AM WEST Phoenix                 415
 8 AM WEST San Diego               448
 9 AM WEST San Francisco           449
10 AM WEST Seattle                 262
```

This is an example of what is commonly described as Simpson's paradox: the relationship observed when data is aggregated can differ from the relationship observed within individual groups. In this dataset, the difference is related to the unequal distribution of flights across destinations.

Conclusion

I started with the original wide-format airline arrival-delay data and recreated it as a CSV file. I used `fill()` to fill in the missing airline names and then used `pivot_longer()` to change the data into a tidy long format. Overall, ALASKA had a delay percentage of about 13.3%, while AM WEST had about 10.9%. However, when I looked at each city separately, ALASKA had a lower delay percentage than AM WEST in all five cities. The difference happens because the airlines have different numbers of flights going to each city. This shows that looking only at the overall percentage can hide differences that are easier to see when the data is broken down by city.